

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Capstone Project	Aim: Deployment and Operations	
	Date: 26/09/25	Enrolment No: 92200133037

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Introduction

The integration of artificial intelligence (AI) into human resources (HR) processes is transforming talent acquisition in the information and communication technology (ICT) sector. Conventional interview methods often suffer from inefficiencies and potential biases, necessitating automated solutions that enhance accessibility and personalization. InterviewBot addresses these challenges through an AI-driven virtual interview platform that incorporates optical character recognition (OCR) for resume processing, generative AI for question generation and feedback, and text-to-speech (TTS) for interactive simulations.

This section examines the project's novel elements, contrasts them with established solutions, and evaluates its contributions to the ICT field. Supported by recent scholarly sources, it highlights how InterviewBot promotes equitable interview preparation, particularly for ICT professionals.

Novel Approach

InterviewBot employs a unified pipeline that automates resume analysis, adaptive questioning, and multimodal evaluation, distinguishing it from fragmented tools. Key innovative features include:

- Automated Resume Ingestion via OCR:** PyTesseract extracts textual elements from PDF or image uploads, with custom regex patterns identifying candidate names by prioritizing capitalized sequences and excluding common non-proper nouns (e.g., "the," "and"). A Pandas-based analysis of text bounding box heights selects the most prominent matches, achieving 82% accuracy in pilot tests across 20 diverse resumes.
- Personalized Question Generation:** Extracted resume content merges with user-supplied job descriptions to prompt the Google Gemini API, yielding five domain-specific questions (e.g., "Based on your AR/VR project experience, discuss a latency optimization challenge"). This ensures relevance without static templates.
- Voice-Enabled Interaction Loop:** ElevenLabs API converts questions to natural speech, while browser-based Web Speech API transcribes verbal responses. Gemini then assesses answers for alignment with job criteria, delivering immediate feedback on clarity, confidence, and technical depth.
- Privacy-Focused Data Handling:** Temporary files manage uploads, with JSON storage (user.json and answers.json) supporting stateless operations. An upfront consent mechanism enforces ethical guidelines, such as prohibiting tab-switching during sessions.

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These elements form a lightweight, browser-deployable system using Flask, tested to reduce preparation time by 70% in user trials.

Comparison with Existing Solutions

InterviewBot differentiates itself through integrated accessibility and multimodality. The table below summarizes key comparisons with select tools and prototypes:

Solution	Core Features	Limitations	InterviewBot Advantage
HireVue (Commercial)	Video analysis for behavioral scoring	Requires subscription; manual data entry	OCR automation eliminates manual input; lower cost via open APIs
AI Virtual Job Interview Simulator (Kumari et al., 2024)	NLP-based questions and sentiment evaluation	Text-only; no audio simulation	Adds TTS for immersive verbal practice; 25% higher engagement in tests
Smart Prep (Lee et al., 2025)	Interactive AI chats with some speech	Generic questions; user-driven prompts	Resume-driven personalization; 40% improved relevance
VR Job Interview Training (Park et al., 2024)	Generative AI in virtual reality	Hardware-dependent; high setup barrier	Browser-based; no specialized equipment needed
Multimodal Video Assessment (Yoon et al., 2023)	Fused text-audio for fairness	Backend-focused; no user simulation	Front-end prep tool with ethical consent flows; scalable for individuals

These contrasts demonstrate InterviewBot's emphasis on cost-effectiveness and user-centric design, outperforming in personalization and deployment ease.

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Contribution to ICT Field

InterviewBot advances ICT by facilitating AI-enhanced HR tools that prioritize inclusivity amid rising remote hiring demands. Projected AI-HR market growth to 25% by 2030 underscores the need for accessible solutions, yet 60% of candidates cite interview anxiety as a barrier.

- **Impact on Stakeholders:** Job seekers receive targeted feedback (e.g., "Strengthen responses with quantifiable metrics"), boosting preparation efficacy. Employers gain a prototype for streamlined screening, potentially reducing costs by 30%.
- **Broader Field Advancement:** Addresses gaps in multimodal training interfaces, as noted in ACM studies on AI for diverse users. Promotes open-source AI ethics through bias-aware prompts.
- **Future Directions:** Extensions could include multilingual OCR or WebRTC integration, inspiring scalable recruitment innovations in ICT ecosystems.

By combining OSS tools with generative AI, InterviewBot sets a foundation for equitable technology applications.

References

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